

STUDYHUB
CLASS 12 — MP BOARD
BIOLOGY — SET 1
ORIGINAL SAMPLE QUESTION PAPER

Time: 3 Hours	Maximum Marks: 70
Medium: English	Set: 1

Note: This is an original StudyHub practice paper based on the MPBSE syllabus and examination pattern. It is not an official MPBSE paper.

GENERAL INSTRUCTIONS

1. All questions are compulsory.
2. Questions 1 to 5 are objective-type questions carrying a total of 28 marks.
3. Questions 6 to 12 carry 2 marks each.
4. Questions 13 to 16 carry 3 marks each.
5. Questions 17 to 20 carry 4 marks each.
6. Internal choices are provided from Question 6 to Question 20.
7. Draw neat and labelled diagrams wherever necessary.
8. Write scientific terms and biological names correctly.

SECTION - A : OBJECTIVE TYPE QUESTIONS

Q.1 Choose the correct option. (1×6=6)

(i) The transfer of pollen grains from anther to stigma of the same flower is called:

- A. Xenogamy
- B. Geitonogamy
- C. Autogamy
- D. Hybridisation

(ii) The hormone responsible for maintaining the uterine lining during pregnancy is mainly:

- A. Insulin
- B. Progesterone
- C. Thyroxine
- D. Adrenaline

(iii) Which molecule carries the genetic information in most organisms?

- A. Protein
- B. Lipid
- C. DNA
- D. ATP

(iv) The enzyme that joins two DNA fragments is:

- A. DNA polymerase
- B. Helicase
- C. DNA ligase
- D. Restriction enzyme

(v) Which of the following is an example of a biological control agent?

- A. *Bacillus thuringiensis*
- B. Urea
- C. DDT
- D. Sodium chloride

(vi) The interaction in which both species are benefited is called:

- A. Predation
- B. Competition
- C. Mutualism

D. Parasitism

Q.2 Fill in the blanks. (1×6=6)

- (i) The fusion of male and female gametes is called _____.
- (ii) The structure that connects the foetus with the placenta is called the _____.
- (iii) The basic unit of heredity is called a _____.
- (iv) The process by which RNA is formed from DNA is called _____.
- (v) The ability of an organism to produce disease is called _____.
- (vi) The maximum population size that an environment can support is called its _____.

Q.3 Write True or False. (1×5=5)

- (i) Spermatogenesis begins at puberty.
- (ii) The placenta acts as an endocrine tissue during pregnancy.
- (iii) RNA generally contains thymine instead of uracil.
- (iv) Natural selection was proposed as an important mechanism of evolution by Darwin.
- (v) All microorganisms are harmful to humans.

Q.4 Match Column A with Column B. (1×6=6)

Column A	Column B
(i) Insulin	(a) Antibody production
(ii) Antibody	(b) Blood glucose regulation
(iii) PCR	(c) DNA amplification
(iv) Bt cotton	(d) Insect resistance
(v) Lichen	(e) Mutualism
(vi) Pyramid of energy	(f) Always upright

Q.5 Answer in one word/one sentence. (1×5=5)

- (i) Name the male reproductive organ that produces sperm.
- (ii) What is the full form of IVF?
- (iii) Who is known as the father of genetics?
- (iv) Name the process by which a mature mRNA is formed from hnRNA.
- (v) What is the term for the movement of individuals into a population?

SECTION - B : VERY SHORT ANSWER QUESTIONS

Q.6 (2 Marks)

What is double fertilisation? Why is it called “double” fertilisation?

OR

Differentiate between self-pollination and cross-pollination.

Q.7 (2 Marks)

What is the menstrual cycle? Mention its approximate duration in humans.

OR

What is contraception? Name any two contraceptive methods.

Q.8 (2 Marks)

What is incomplete dominance? Give one example.

OR

State Mendel's law of segregation.

Q.9 (2 Marks)

What is a codon? State any one important property of the genetic code.

OR

What is transcription? Where does it occur in a eukaryotic cell?

Q.10 (2 Marks)

What is vaccination? How does it help in developing immunity?

OR

What is an antibiotic? Name one antibiotic obtained from a microorganism.

Q.11 (2 Marks)

What is recombinant DNA technology?

OR

What are restriction enzymes? Why are they important in genetic engineering?

Q.12 (2 Marks)

Define ecological succession.

OR

What is biodiversity? Mention one major cause of biodiversity loss.

SECTION - C : SHORT ANSWER QUESTIONS

Q.13 (3 Marks)

Describe the structure of a mature human sperm with a labelled diagram.

OR

Explain the process of oogenesis in humans.

Q.14 (3 Marks)

Explain Mendel's monohybrid cross and state the phenotypic ratio obtained in the F₂ generation.

OR

Explain the inheritance of ABO blood groups in humans.

Q.15 (3 Marks)

Explain the process of DNA replication. Mention the role of DNA polymerase.

OR

Describe the main steps involved in translation.

Q.16 (3 Marks)

Explain the principle and major steps of Polymerase Chain Reaction (PCR).

OR

Explain how insulin is produced using recombinant DNA technology.

SECTION - D : LONG ANSWER QUESTIONS

Q.17 (4 Marks)

Describe the structure of the human female reproductive system with a neat labelled diagram.

OR

Explain the major events of the menstrual cycle and mention the role of FSH, LH, estrogen and progesterone.

Q.18 (4 Marks)

Explain the chromosomal theory of inheritance. How did it provide a physical basis for Mendel's laws?

OR

Explain sex determination in humans using a suitable cross.

Q.19 (4 Marks)

Explain the process of evolution by natural selection. Mention the importance of variations in evolution.

OR

Explain Hardy-Weinberg principle and write any two factors that can disturb genetic equilibrium.

Q.20 (4 Marks)

Explain the role of microorganisms in human welfare with suitable examples from: (i) Food production (ii) Industrial production (iii) Sewage treatment (iv) Biogas production.

OR

Explain the importance of biodiversity and describe any two methods used for its conservation.

MARKS VERIFICATION

Questions	Marks
Q.1	6
Q.2	6
Q.3	5
Q.4	6
Q.5	5
Q.6-Q.12	14
Q.13-Q.16	12
Q.17-Q.20	16
TOTAL	70

6 + 6 + 5 + 6 + 5 + 14 + 12 + 16 = 70 Marks ✓

SHORT ANSWER KEY

Q.1: C, B, C, C, A, C. Q.2: fertilisation; umbilical cord; gene; transcription; pathogenicity; carrying capacity. Q.3: True, True, False, True, False. Q.4: (i)-(b), (ii)-(a), (iii)-(c), (iv)-(d), (v)-(e), (vi)-(f). Q.5: Testis; In Vitro Fertilisation; Gregor Johann Mendel; RNA processing/splicing; immigration.

Q.8: Monohybrid F_2 phenotypic ratio = 3:1. Q.14: ABO blood group inheritance involves I^A , I^B and i alleles. Q.18: XX = female, XY = male. Q.19: $p^2 + 2pq + q^2 = 1$. For Q.20, award marks for appropriate examples and biological explanations. Full solutions are not included.